# Maxwell Rosen

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## Education

### **University of California, San Diego, CA** - *Bachelor of Science*

September 2024 - Expected June 2028

- Chemical Engineering Major | Business Economics Minor
- **GPA: 3.96**
- Relevant Coursework: Chemical Reaction Engineering, Thermodynamics, Material and Energy Balances, Organic Chemistry
- Marshall College Honors Program

## Experience

### **Envision Maker Studio** - *Volunteer*

UC San Diego • October 2026 - Present

- In the campus makerspace, troubleshooting and guiding safe usage of 3D printers, laser cutters, and fabrication tools for peers' personal and course projects

### **Biomedical Device Research** - *Visiting Undergraduate Researcher*

National Taiwan University • June 2026 - July 2026

- Established saturation baselines by mass and imaging to set coating parameters and optimize Western blot experiments
- Developed an automated micropipetting module that eliminated manual reagent dispensing for Western blot and IHC
- Identified a 50% reduction in post-coating vacuum time (10 s to 5 s) with no loss in signal; presented findings to NTU faculty and students

### **Salk Institute for Biological Studies Research** - *Undergraduate Research Intern*

Ecker Lab • September 2025 - September 2026

- Designed and deployed 4 Raspberry Pi–based high-throughput imaging systems at 1% the cost of commercial alternatives ($500 vs. $50,000 per unit) to monitor plant growth across 200+ samples under variable environmental stress
- Doubled imaging pipeline capacity from 4 to 8 trays and integrated scalability by resolving hardware-software integration issues on a 4-person team
- Documented system architecture and authored setup protocols

### **Lighting Technician** - *Utility*

ShowPro, Freelance • September 2025 - Present

- Execute live event production through fixture placement, cable routing, dimmer/DMX setup, and power distribution to meet show specifications and ensure efficient run-of-show operations

## Publications

Ovadia R, **Rosen MG**, Li KR, Abhilashkumar AV, Toda TJ, Jow B, Ecker JR, Illouz-Eliaz N. "PhenOmni: A Low-Cost, Modular Phenotyping Platform Integrating Automated Imaging and Soil Moisture Sensing for High-Throughput *Arabidopsis thaliana* Growth Analysis." *Plants* (MDPI), under revision, 2026.

## Skills

- Programming (Python, Git, MATLAB, Excel)
- Hardware (Raspberry Pi, ESP32)
- CAD (Fusion)
- Lab (Western blot, IHC, imaging, data analysis)

## Certifications

- AIChE SAChE: CCPS Process Safety Fundamentals In Progress